

Miranda Mai

ENGLISH, MANDARIN, CANTONESE

AI & MARKETING ENTHUSIAST

Proficient in Python, SQL, HTML/CSS and C++. Experienced in building and fine-tuning machine learning models, including optimizing hyperparameters to improve model performances. Skilled in using tools such as Google Colab, Microsoft Azure, and Databricks for data processing and model development.

Expert Skills

Data Processing & Analysis ● ● ● ● ●

Machine Learning Skills ● ● ● ● ●

Marketing & Design Skills ● ● ● ● ●

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[Portfolio Link](#)

Central, Auckland

Education History

MASSEY UNIVERSITY

- Master of Information Science, 2023-2025
- Core Courses: Data Science, Machine Learning, Computer Vision
- Conducted fine-tuning large language models (LLMs) for a one-year research project
- GPA: 6.5/9

MASSEY UNIVERSITY

- Master of Management (Marketing), 2019-2020
- GPA: 6.92/9
- Class Advocate
- VP (Pub Relationship) of Toastmasters

Work Experience

MARKETING COORDINATOR

Fletcher Building Ltd | Mar 2022 - July 2023

- Website management and maintenance
- Organized conferences, managed merchant displays and merchandising, and oversaw budgeting for marketing activities
- Assisted with managing social media channels to promote sales and measure results
- Conducted marketing research to support strategy and brand management

MARKETING COORDINATOR

NZ Soap & Skincare Ltd | May - Dec 2020

- Developed buyer personas localized for specific markets
- Assisted in creating marketing strategies and brand management
- Managed social media channels to promote sales
- Organized project plans for campaigns and measured results
- Website development and maintenance

INVENTORY MANAGER

Netpharmacy | Oct 2016 - Dec 2018

- Conducted inventory control, stock-taking and retail processing
- Supervised dispatch team and trained new staff
- Used the ISAM system for parcel dispatch and shipping
- Coordinated across departments to ensure timely online order dispatch

Portfolio

About Me

Data Science professional with a Master's degree in Information Sciences, specializing in end-to-end data processing and machine learning workflows. I possess hands-on experience in fine-tuning models and implementing advanced algorithms to derive actionable insights from complex datasets. By leveraging a strong technical foundation in website development and data architecture, I bridge the gap between raw data and scalable, intelligent solutions that drive measurable growth.

Featured Projects

1. AI & LLM Research: Error Detection in Education

Tech Stack: Python, Transformers, NLP, Llama-3.2, Mixtral-8x7B, Deepseek, Prompt Engineering, Model Fine-tuning.

Key Contributions:

- **Benchmarked Reliability:** Analyzed 5,000+ Math Word Problems (MWP) across 12 error categories to identify AI safety gaps in education.
- **Optimized Framework:** Achieved 90% Zero-Shot accuracy using Llama-3.2; proved that "prompt minimalism" outperforms complex prompting in specific domains.
- **Bias Analysis:** Identified pretraining biases and output inconsistencies, providing a roadmap for adversarial fine-tuning and data-quality improvements.

Artifacts:

- [Full Thesis](#) [GitHub Repo](#)

2. Project: The News Trend Predictor – AI-Driven Content Strategy Tool

Technical Stack: Python (PyTrends), YouTube Data API, Machine Learning, Web App Development (Streamlit/Flask), Data Wrangling.

Core Achievements:

- **Predictive Engine:** Built a web app forecasting Google Search Trends (0-100 scale) to predict "virality trajectories" for SEO and marketing insights.
- **Data Pipeline:** Engineered a custom pipeline aggregating Google Trends and YouTube metrics for holistic content relevancy analysis.
- **Model Optimization:** Benchmarked ML algorithms to identify optimal forecasting models, improving reliability through robust feature selection.

Artifacts:

- [GitHub Repo](#)

3. Full-Stack Web Development & Deployment

Tech Stack: JavaScript, HTML5/CSS3, Python, Flask, Jinja, MySQL.

Core Achievements:

- **End-to-End Delivery:** Designed and deployed multiple responsive web apps, integrating dynamic front-ends with robust MySQL back-end logic.
- **Architecture & DevOps:** Managed the full lifecycle from database schema design to cloud deployment on PythonAnywhere, ensuring high availability.
- **Live Impact:** Delivered a Professional Services Platform and a Veterinary Data Management System.

Live Projects:

- [Professional Services Platform](#)
- [Veterinary Services Web with Data Management System](#)

4. Project: Data-Driven Marketing Engine – Multi-Channel Growth Strategy

Technical Stack: TikTok/LinkedIn/WeChat Analytics, SEO/SEM, Content Performance Tracking, Data-Driven Engagement Optimization.

Core Achievements:

- **Exponential Growth:** Scaled a TikTok account to 10,000+ followers and 110,000+ likes in 60 days via high-frequency data testing.
- **Rapid Acquisition:** Achieved 1,000+ WeChat followers in 10 days through targeted segment analysis and content positioning.
- **Strategy Optimization:** Leveraged cross-platform data to increase organic engagement by 300% and unify the brand ecosystem.

Live Channels:

- TikTok Channel — High-growth short-video platform on [ID: [MiLanDa0505](#)].
- WeChat Video Channel — Rapid-growth professional content hub, channel ID: MaxMai

Connect with Me

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- GitHub: <https://github.com/MirandaZmai>

